# GET SET HIRED

DING ROUND • PATTERNS • LEETCODE • CODING ROUND • PATTERNS • LEETCODE • CODING ROUND • PATTERNS

EP 01

## 15 Coding Patterns · 75 Problems

Stop solving random questions. Learn the pattern once, and you can solve problems you've never seen.

# HOW TO USE THIS

**1**

### One pattern at a time

Read the idea and the visual first. Then solve its 5 problems in order, easy to hard.

**2**

### Timebox 30 min

Stuck after 30 minutes? Read the solution, understand it, then re-solve it yourself the next day.

**3**

### Say it out loud

Before coding, explain the approach and its time complexity, like you would to an interviewer.

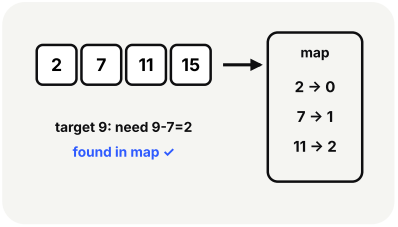
### Suggested 30-day plan

| | |
|---|---|
| Days 1–10 | Patterns 01–05 · arrays, pointers, windows, prefix sums, binary search |
| Days 11–20 | Patterns 06–11 · linked lists, stacks, heaps, intervals, trees |
| Days 21–30 | Patterns 12–15 · graphs, backtracking, dynamic programming + revision |

## Which pattern? Quick decoder

| IF THE QUESTION SAYS... | TRY |
|---|---|
| "sorted array", "pair that sums to" | Two Pointers / Binary Search |
| "longest / shortest substring or subarray" | Sliding Window |
| "sum of subarray equals K" | Prefix Sum + Hash Map |
| "top K", "Kth largest", "median" | Heap |
| "next greater", "valid brackets" | Stack |
| "overlapping", "meetings", "ranges" | Intervals |
| "all combinations / permutations / subsets" | Backtracking |
| "number of ways", "minimum cost", "can you reach" | Dynamic Programming |
| "grid", "islands", "prerequisites", "shortest path" | Graph BFS / DFS |

# 01 Arrays & Hashing

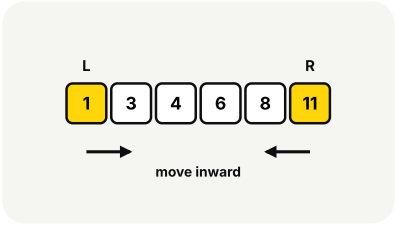



Use a hash map/set to get O(1) lookups instead of nested loops.

**Spot it when:** Need to find duplicates, counts, pairs or groups fast.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #1 | Two Sum | Easy |
| ☐ | #217 | Contains Duplicate | Easy |
| ☐ | #242 | Valid Anagram | Easy |
| ☐ | #49 | Group Anagrams | Med |
| ☐ | #347 | Top K Frequent Elements | Med |

# 02 Two Pointers

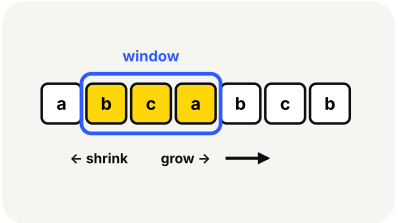



Two indexes move toward each other (or together) over a sorted array/string.

**Spot it when:** Sorted input, pairs/triplets with a target sum, palindromes.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #125 | Valid Palindrome | Easy |
| ☐ | #167 | Two Sum II – Input Array Is Sorted | Med |
| ☐ | #15 | 3Sum | Med |
| ☐ | #11 | Container With Most Water | Med |
| ☐ | #42 | Trapping Rain Water | Hard |

# 03 Sliding Window




Grow the right edge, shrink the left edge while a condition breaks.

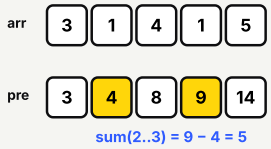
**Spot it when:** Longest/shortest subarray or substring that satisfies a condition.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #121 | Best Time to Buy and Sell Stock | Easy |
| ☐ | #3 | Longest Substring Without Repeating Characters | Med |
| ☐ | #424 | Longest Repeating Character Replacement | Med |
| ☐ | #567 | Permutation in String | Med |
| ☐ | #76 | Minimum Window Substring | Hard |

## 04 Prefix Sum

Precompute running totals so any range sum is one subtraction.

**Spot it when:** Range sums, subarray sum equals K, balance problems.




| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #724 | Find Pivot Index | Easy |
| ☐ | #303 | Range Sum Query – Immutable | Easy |
| ☐ | #238 | Product of Array Except Self | Med |
| ☐ | #560 | Subarray Sum Equals K | Med |
| ☐ | #525 | Contiguous Array | Med |

## 05 Binary Search

Halve the search space each step, on the array or on the answer.

**Spot it when:** Sorted data, rotated arrays, 'minimum X such that...' questions.

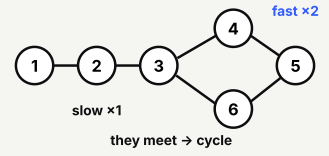



| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #704 | Binary Search | Easy |
| ☐ | #74 | Search a 2D Matrix | Med |
| ☐ | #875 | Koko Eating Bananas | Med |
| ☐ | #153 | Find Minimum in Rotated Sorted Array | Med |
| ☐ | #33 | Search in Rotated Sorted Array | Med |

## 06 Fast & Slow Pointers

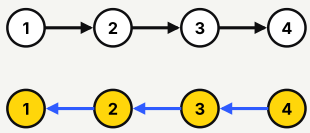
One pointer moves 1 step, the other 2. If there's a cycle, they meet.

**Spot it when:** Cycle detection, middle of a list, repeated sequences.




| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #141 | Linked List Cycle | Easy |
| ☐ | #876 | Middle of the Linked List | Easy |
| ☐ | #202 | Happy Number | Easy |
| ☐ | #143 | Reorder List | Med |
| ☐ | #287 | Find the Duplicate Number | Med |

## 07 Linked List Reversal

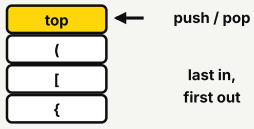



Re-point next pointers in place using prev / curr / next.

**Spot it when:** Reverse a whole list, part of a list, or in groups.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #206 | Reverse Linked List | Easy |
| ☐ | #21 | Merge Two Sorted Lists | Easy |
| ☐ | #19 | Remove Nth Node From End of List | Med |
| ☐ | #2 | Add Two Numbers | Med |
| ☐ | #25 | Reverse Nodes in k-Group | Hard |

## 08 Stack & Monotonic Stack

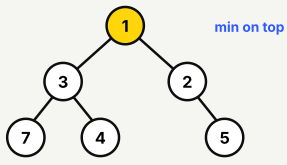



Push/pop to match pairs; keep the stack sorted to find next greater/smaller.

**Spot it when:** Brackets, expression evaluation, 'next warmer day' type questions.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #20 | Valid Parentheses | Easy |
| ☐ | #155 | Min Stack | Med |
| ☐ | #150 | Evaluate Reverse Polish Notation | Med |
| ☐ | #739 | Daily Temperatures | Med |
| ☐ | #84 | Largest Rectangle in Histogram | Hard |

## 09 Heap / Top K

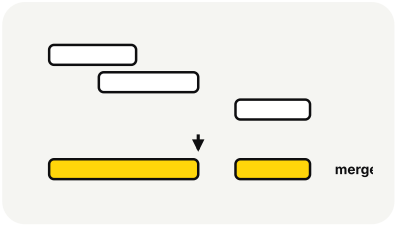



Keep a min-heap of size K so the top is always the Kth best.

**Spot it when:** Top K, Kth largest, running median, merging sorted streams.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #703 | Kth Largest Element in a Stream | Easy |
| ☐ | #1046 | Last Stone Weight | Easy |
| ☐ | #215 | Kth Largest Element in an Array | Med |
| ☐ | #973 | K Closest Points to Origin | Med |
| ☐ | #295 | Find Median from Data Stream | Hard |

## 10 Intervals

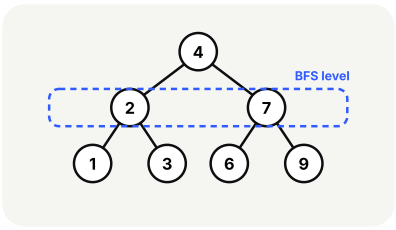



Sort by start time, then merge or compare neighbours.

**Spot it when:** Meetings, overlapping ranges, scheduling.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #56 | Merge Intervals | Med |
| ☐ | #57 | Insert Interval | Med |
| ☐ | #435 | Non-overlapping Intervals | Med |
| ☐ | #452 | Minimum Number of Arrows to Burst Balloons | Med |
| ☐ | #1851 | Minimum Interval to Include Each Query | Hard |

## 11 Tree DFS & BFS

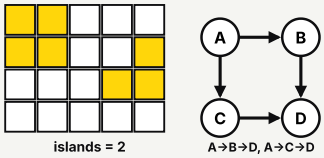



Recursion (DFS) for depth/paths, a queue (BFS) for level by level.

**Spot it when:** Any binary tree or BST question.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #226 | Invert Binary Tree | Easy |
| ☐ | #104 | Maximum Depth of Binary Tree | Easy |
| ☐ | #102 | Binary Tree Level Order Traversal | Med |
| ☐ | #98 | Validate Binary Search Tree | Med |
| ☐ | #124 | Binary Tree Maximum Path Sum | Hard |

## 12 Graph BFS / DFS / Topological Sort

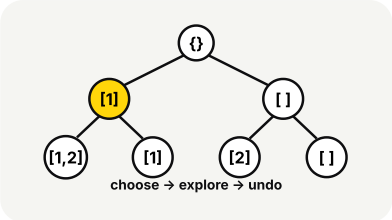



Visit neighbours with a visited set; topo sort for dependencies.

**Spot it when:** Grids, islands, networks, course prerequisites.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #200 | Number of Islands | Med |
| ☐ | #133 | Clone Graph | Med |
| ☐ | #994 | Rotting Oranges | Med |
| ☐ | #207 | Course Schedule | Med |
| ☐ | #127 | Word Ladder | Hard |

## 13 Backtracking

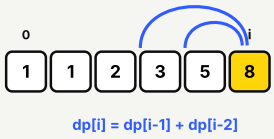



Choose → explore → un-choose. Try every option, prune bad branches.

**Spot it when:** Generate all subsets, permutations, combinations, board puzzles.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #78 | Subsets | Med |
| ☐ | #46 | Permutations | Med |
| ☐ | #39 | Combination Sum | Med |
| ☐ | #79 | Word Search | Med |
| ☐ | #51 | N-Queens | Hard |

## 14 1D Dynamic Programming

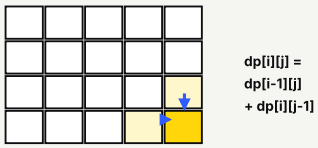



Answer for i built from answers for i-1, i-2... Store them in an array.

**Spot it when:** 'How many ways', 'min/max cost', choices at each step.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #70 | Climbing Stairs | Easy |
| ☐ | #198 | House Robber | Med |
| ☐ | #322 | Coin Change | Med |
| ☐ | #300 | Longest Increasing Subsequence | Med |
| ☐ | #139 | Word Break | Med |

## 15 2D Dynamic Programming




A grid dp[i][j] built from the cell above / left / diagonal.

**Spot it when:** Two strings, grids, knapsack-style choices.

| | NO. | PROBLEM | LEVEL |
|---|---|---|---|
| ☐ | #62 | Unique Paths | Med |
| ☐ | #1143 | Longest Common Subsequence | Med |
| ☐ | #518 | Coin Change II | Med |
| ☐ | #72 | Edit Distance | Med |
| ☐ | #312 | Burst Balloons | Hard |

# PRACTICE FROM THESE 3 SHEETS

Done with the 75 above? Move to a full sheet. Pick one, not all three.

## Blind 75

The original 75 must-do problems. Best if you have 4–6 weeks.

neetcode.io/practice → Blind 75 tab

## NeetCode 150

Blind 75 plus 75 more, grouped by pattern, with a video solution for every problem.

neetcode.io/practice → NeetCode 150 tab

## Striver's SDE Sheet (now "Striver's 180")

Revamped as 179 problems across 17 DSA pattern modules, with articles and videos.

takeuforward.org · Striver's SDE Sheet

**Up next · EP 02**

Technical rounds: the OS, DBMS, CN and OOPs questions they always ask. Follow **@techie_programmer** so you don't miss it.